

SAFETY DATA SHEET (GHS)

DMBA — 2-Methyl-1-phenyl-2-propanol (Dimethylbenzylcarbinol) | CAS: 100-86-7 | Formula: C₁₀H₁₄O

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ChemicalBook — 2-Methyl-1-phenyl-2-propanol, CAS 100-86-7, <https://www.chemicalbook.com/>
- TCI America — Product D0783, 2-Methyl-1-phenyl-2-propanol (CAS 100-86-7), <https://www.tcichemicals.com/>
- PubChem — CAS 100-86-7 identity cross-check, <https://pubchem.ncbi.nlm.nih.gov/>
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: DMBA (2-Methyl-1-phenyl-2-propanol / Dimethylbenzylcarbinol)

CAS Number: 100-86-7

Chemical formula: C₁₀H₁₄O

Molecular weight: 150.22 g/mol

Synonyms: 2-Methyl-1-phenyl-2-propanol; Benzyl dimethylcarbinol; Dimethylbenzyl carbinol (DMBC)

IDENTITY VERIFICATION NOTE: The HAZOP PSI brief refers to this stream as 'DMBA' — in cumene-process literature, 'DMBA' most commonly denotes alpha,alpha-dimethylbenzyl alcohol (2-phenyl-2-propanol, PhC(CH₃)₂OH, C₉H₁₂O, CAS 617-94-7), the direct alcohol intermediate formed during CHP decomposition (which dehydrates to AMS, CAS 98-83-9, the chemical immediately preceding this one in the brief's list). However, the CAS number specified in the brief (100-86-7) corresponds to a DIFFERENT, structurally related isomer: 2-Methyl-1-phenyl-2-propanol (PhCH₂C(CH₃)₂OH, C₁₀H₁₄O), also colloquially called 'dimethylbenzylcarbinol'. This SDS has been compiled for CAS 100-86-7 AS SPECIFIED in the brief. RECOMMEND: confirm with PPCL process engineering which specific isomer/CAS number corresponds to the actual DMBA stream in Train II (i.e., CAS 100-86-7 vs. CAS 617-94-7) — this is analogous in nature to the Diamine/TBC identity-confirmation item (CAS 110-97-4 placeholder) and should be resolved before this SDS is relied upon for PSI purposes.

Process role: Intermediate alcohol associated with CHP decomposition chemistry (dehydrates to AMS-type olefins under acid catalysis); see Identity Verification Note above. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Acute Tox. 4 (H302) — based on available supplier hazard statement data for CAS 100-86-7; some sources additionally cite H301 (Acute Tox. 3, Toxic if swallowed) — see Other Hazards note below

Signal word: Warning

Hazard statements (H-codes): H302 (Harmful if swallowed) — primary classification used in this SDS

Precautionary statements (P-codes): P264, P270, P301+P312, P330, P501

GHS pictograms: GHS07 (Exclamation mark)

Other hazards: DATA-QUALITY FLAG: search results returned conflicting acute oral toxicity hazard statements for this CAS number — both H301 (Toxic if swallowed, Acute Tox. 3) and H302 (Harmful if swallowed, Acute Tox. 4) appear in different source listings. This SDS conservatively notes both possibilities but uses H302/Acute Tox. 4 with Warning as the primary classification pending confirmation. RECOMMEND verifying against the specific supplier SDS used on-site — if H301/Acute Tox. 3 is confirmed, the signal word would change to 'Danger'. See also the Identity Verification Note in Section 1 regarding possible CAS/isomer discrepancy.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Substance: 2-Methyl-1-phenyl-2-propanol (or alpha,alpha-dimethylbenzyl alcohol — see Section 1 identity note), as encountered in process intermediate streams; purity/composition to be confirmed against PPCL process data

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air. Obtain medical attention if symptoms persist.

Skin contact: Wash with plenty of soap and water. Remove contaminated clothing.

Eye contact: Flush with copious water for at least 15 minutes. Obtain medical attention if irritation persists.

Ingestion: Rinse mouth. Do not induce vomiting. Obtain medical attention immediately (H302; treat as potentially H301 pending confirmation per Section 2).

Notes to physician: Treat symptomatically; given the acute oral toxicity flag in Section 2, monitor for systemic effects following significant ingestion

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Dry chemical, CO₂, alcohol-resistant foam, water spray (for cooling)

Unsuitable extinguishing media: None specific

Specific hazards: This material is a solid/low-melting solid near room temperature (melting point ~24-26°C reported for the 2-methyl-1-phenyl-2-propanol isomer) with a relatively high flash point — primary fire hazard is limited under ambient conditions but should be assessed against the actual process stream composition and temperature

Special protective equipment for firefighters: Self-contained breathing apparatus (SCBA) and full protective clothing for fires involving this material

Hazardous combustion products: Carbon monoxide, carbon dioxide, smoke

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Avoid ingestion and prolonged skin contact; wear appropriate PPE (Section 8)

Environmental precautions: Avoid release to drains/watercourses pending confirmation of ecological hazard classification

Containment and cleanup: Contain spill/solid residue; collect into closed, labelled containers for disposal; if molten/liquid (above ~25°C), treat as a viscous liquid spill

Reference to other sections: See Section 7 (Handling/Storage)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Avoid ingestion, prolonged skin contact, and inhalation of vapours/mist; this material may be solid at typical ambient temperatures (MP ~24-26°C) — handling systems may require gentle heat tracing to maintain flow if handled as a liquid

Storage conditions: Store in tightly closed containers in a cool, dry, well-ventilated area

Temperature limits: Note proximity of melting point (~24-26°C) to ambient temperature — storage/transfer temperature should be controlled to maintain the desired phase (solid vs. liquid) as required by the process

Pressure: Atmospheric storage

Segregation requirements: Segregate from strong oxidisers and strong acids

Special requirements: None beyond general good practice pending confirmation of identity (Section 1) and full hazard profile

Specific end-use recommendations: PRIORITY ACTION: confirm actual chemical identity/CAS number for the 'DMBA' process stream (Section 1) before finalising storage/handling procedures based on this SDS

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): No specific ACGIH TLV or OSHA PEL established for this CAS number; confirm Thailand OEL if applicable

TWA: Not established

STEL: Not established

Ceiling: Not established

Skin notation: Not assigned

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen

Engineering controls: General ventilation adequate for typical handling; local exhaust for tasks generating vapour/mist or dust (if handled as solid)

PPE — Respiratory: Not normally required for typical handling; organic vapour respirator if vapour/mist concentrations are elevated

PPE — Gloves: Chemical-resistant gloves (e.g., nitrile)

PPE — Eyes: Safety glasses or chemical splash goggles

PPE — Body: Standard work clothing; chemical-resistant apron for bulk transfer

Biological monitoring: Not applicable

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: White to off-white crystalline solid or viscous liquid near ambient temperature

Odour: Mild, characteristic odour

Odour threshold: Not established

pH: Not applicable

Melting/Freezing point: ~24-26°C (2-methyl-1-phenyl-2-propanol isomer) — confirm against the actual process stream identity (see Section 1)

Boiling point (at 1 atm): ~217°C (2-methyl-1-phenyl-2-propanol isomer; the alternative alpha,alpha-dimethylbenzyl alcohol isomer, CAS 617-94-7, has a reported boiling point of ~202-205°C — confirm against actual stream identity)

Flash point: ~92-96°C (estimated from related dimethylbenzyl alcohol isomers; confirm against supplier SDS for actual stream)

Evaporation rate: Slower than water

Flammability (solid/gas): Combustible solid/liquid; not classified as readily flammable

Flammability limits in air: Not established

Vapour pressure (at 20°C): Low — material has low volatility at ambient temperature

Vapour density (air = 1): ~5.2 (estimated from molecular weight)

Relative density (water = 1): ~0.98-1.0 (estimated)

Solubility (water): Slightly soluble

Partition coefficient (log K_{ow}): ~1.7-2.0 (estimated)

Auto-ignition temperature: Not established

Decomposition temperature: Not established under normal conditions; tertiary benzylic alcohols of this type can undergo acid-catalysed dehydration to the corresponding alkene (e.g., to AMS-type olefins) under process conditions — relevant to the process chemistry context

Viscosity: Not established

Explosive properties: Not classified as explosive

Oxidising properties: Not an oxidiser

Molecular weight: 150.22 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal conditions; tertiary benzylic alcohols of this class can undergo acid-catalysed dehydration to form alkenes (e.g., AMS-type olefins) — this is consistent with this material's role as a process intermediate

Chemical stability: Stable under recommended storage conditions

Possibility of hazardous reactions: Dehydration to olefin under strong acid catalysis (relevant given proximity to sulfuric acid, CAS 7664-93-9, in the process); no other particular hazardous reactions known

Conditions to avoid: Contact with strong acids/dehydrating agents outside of the intended process step; heat

Incompatible materials: Strong acids, strong oxidisers

Hazardous decomposition products: Carbon monoxide, carbon dioxide on combustion; alkene formation (e.g., AMS) under acid-catalysed dehydration

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Ingestion, inhalation, skin/eye contact

Acute toxicity — Oral LD₅₀: Not precisely established for this CAS number from available sources — classified Acute Tox. 4 (H302) per available hazard statement data, with a possible Acute Tox. 3 (H301) discrepancy noted in Section 2; confirm

against supplier SDS

Acute toxicity — Dermal LD50: Not established

Acute toxicity — Inhalation LC50: Not established

Skin corrosion/irritation: Not classified as corrosive based on available data; may cause mild irritation

Serious eye damage/irritation: Not classified under the primary classification used here; confirm against supplier SDS

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Not classified

Carcinogenicity: Not classified as a carcinogen

Reproductive toxicity: Not classified

STOT (single exposure): Not classified

STOT (repeated exposure): Not classified

Aspiration hazard: Not established

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Not well established for this CAS number from available sources

Persistence and degradability: Not established; benzylic alcohols of this type are generally expected to be biodegradable

Bioaccumulative potential: Low to moderate (estimated log Kow ~1.7-2.0)

Mobility in soil: Not established

Other adverse effects: None specifically identified

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via licensed hazardous waste incineration or treatment facility in accordance with local regulations; do not discharge to drains

Contaminated packaging: Dispose of as hazardous waste or, if uncontaminated/empty, per local regulations after appropriate purging

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535 (verify applicability); local industrial waste regulations

SECTION 14 — TRANSPORT INFORMATION

UN Number: Not regulated as a dangerous good for transport based on available data (confirm for the specific stream/identity used on-site)

Proper shipping name: Not applicable / Not regulated

Transport hazard class(es): Not classified as a transport hazard based on available data

Packing group: Not applicable

Environmental hazards (marine pollutant): Not classified as a marine pollutant based on available data

Special precautions for user: None specific identified

Bulk transport (MARPOL/IBC): Not established — confirm if applicable

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 100-86-7). IDENTITY FLAG: possible discrepancy between the CAS number specified in the brief (100-86-7, 2-methyl-1-phenyl-2-propanol) and the substance most commonly referred to as 'DMBA' in cumene-process literature (alpha,alpha-dimethylbenzyl alcohol, CAS 617-94-7) — see Section 1. ACUTE TOXICITY FLAG: conflicting H301/H302 hazard statements found for CAS 100-86-7 — see Section 2. Both flags require manual verification before this SDS is relied upon for PSI purposes.

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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